

Project 01 — Bluetooth & WiFi Controlled Car



ROBOTICS LAB

BEGINNER PROJECT

ESP32 + ARDUINO

This project workbook walks you through building **two versions** of a remote-controlled car — one using **Bluetooth (HC-05 + Arduino)** and another using **WiFi (ESP32 Web Server)**. You'll learn wiring, motor control, wireless communication, and code logic — all in a hands-on, beginner-friendly format. Think of this as your lab notebook companion!

Objective & What You'll Build

Project Goal

Build a small RC car that you can control from your smartphone — no physical remote needed! The car receives wireless commands and moves accordingly.

- Forward, Backward, Left, Right movement
- Smartphone as the controller (Bluetooth or Web)
- Two versions: Arduino + HC-05 AND ESP32 + WiFi

Skills You'll Learn

- Serial communication (UART)
- Motor driver circuits (L298N)
- Bluetooth module pairing
- ESP32 WiFi server setup
- Reading and writing motor control code

 ★ Both versions use the same chassis and motor driver — only the communication module changes!



Components Required

Gather all components before you start. Check each item off as you collect it. Most of these are available at local electronics shops or online (Amazon, Robu.in, ElectronicsComp).



Microcontroller

- Arduino Uno (Bluetooth version)
- ESP32 DevKit (WiFi version)



Communication Module

- HC-05 Bluetooth Module
- ESP32 has WiFi built-in ✓



Motor Driver

- L298N Motor Driver Module
- Handles 2 DC motors easily



Power & Chassis

- 2× 18650 Li-ion batteries (7.4V)
- Car chassis kit with 2 DC motors
- Jumper wires, breadboard



Tip: Get a pre-assembled car chassis kit — it saves a LOT of time on mechanical assembly!

Version 1 — Bluetooth Car Circuit

Here's how everything connects for the **Arduino + HC-05** version. Study this wiring table carefully before touching a single wire!

Component A	Pin A	Component B	Pin B
Arduino Uno	TX (Pin 1)	HC-05 Bluetooth	RX
Arduino Uno	RX (Pin 0)	HC-05 Bluetooth	TX
Arduino Uno	5V	HC-05 / L298N	VCC
Arduino Uno	GND	HC-05 / L298N	GND
Arduino Pin 5	Digital OUT	L298N	IN1
Arduino Pin 6	Digital OUT	L298N	IN2
Arduino Pin 9	Digital OUT	L298N	IN3
Arduino Pin 10	Digital OUT	L298N	IN4
Battery Pack (+)	7.4V	L298N	12V Power Input
Left Motor	Wire 1 & 2	L298N	OUT1 & OUT2
Right Motor	Wire 1 & 2	L298N	OUT3 & OUT4

Smartphone

HC-05 Module

Arduino Uno

L298N Motor Driver

Signal flows from your phone → Bluetooth module → Arduino → Motor Driver → Motors. Simple chain! ⚡



Bluetooth Module — HC-05

What is the HC-05?

The HC-05 is a **serial Bluetooth module** that acts as a wireless bridge. It receives data from your phone and sends it to Arduino via UART (serial).

- Works at **9600 baud** by default
- Range: up to **10 meters**
- Power: **3.3V – 5V**
- Pairs easily with Android apps



Default password: **1234** or **0000**

HC-05 Pinout

- **VCC** → Arduino 5V
- **GND** → Arduino GND
- **TX** → Arduino RX (Pin 0)
- **RX** → Arduino TX (Pin 1)
- **STATE** → Optional (LED indicator)
- **EN** → Optional (enable pin)



Important: Disconnect HC-05 TX/RX from Arduino before uploading code, then reconnect after upload!

Motor Driver — L298N

The L298N is the **muscle** of your car. Arduino can't power motors directly (too much current!), so the L298N acts as a power amplifier.

IN1 & IN2

Controls **Left Motor** direction. Arduino Pins 5 & 6.

IN3 & IN4

Controls **Right Motor** direction. Arduino Pins 9 & 10.

OUT1-OUT4

Connect your **DC motors** here (2 wires per motor).

12V Power

Connect **battery pack** here. Jumper ON for 5V logic.

Motor Truth Table — How Direction Works

IN1	IN2	IN3	IN4	Car Moves
HIGH	LOW	HIGH	LOW	⬆ Forward
LOW	HIGH	LOW	HIGH	⬇ Backward
LOW	HIGH	HIGH	LOW	↶ Left
HIGH	LOW	LOW	HIGH	↷ Right
LOW	LOW	LOW	LOW	◻ Stop

Control Commands — How Your Phone Talks to the Car

Your phone app (like **Arduino Bluetooth Controller**) sends single-character commands. Arduino reads them and moves the car accordingly.

Key	Command	Movement	Motor State
 F	'F'	Forward	Both motors forward
 B	'B'	Backward	Both motors reverse
 L	'L'	Turn Left	Left stop, Right forward
 R	'R'	Turn Right	Left forward, Right stop
 S	'S'	Stop	All motors off

Recommended App

Arduino Bluetooth Controller by Gaurav (Android)

- Free on Play Store
- Custom button layout
- Sends F, B, L, R, S characters

Pairing Steps

1. Power on HC-05 (LED blinks)
2. Open phone Bluetooth settings
3. Pair with "HC-05", password **1234**
4. Open app → select HC-05
5. Press buttons → car moves!



Arduino Code Logic — Bluetooth Version

Here's the simplified Arduino sketch. Upload this to your Arduino Uno (disconnect HC-05 TX/RX first!).

```
#include <SoftwareSerial.h>

// HC-05 on pins 2 (RX) and 3 (TX)
SoftwareSerial bt(2, 3);

// Motor pins
int in1 = 5, in2 = 6; // Left motor
int in3 = 9, in4 = 10; // Right motor

void setup() {
  bt.begin(9600); // Bluetooth baud rate
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
}

void forward() { digitalWrite(in1,HIGH); digitalWrite(in2,LOW);
                 digitalWrite(in3,HIGH); digitalWrite(in4,LOW); }
void backward() { digitalWrite(in1,LOW); digitalWrite(in2,HIGH);
                 digitalWrite(in3,LOW); digitalWrite(in4,HIGH); }
void left()     { digitalWrite(in1,LOW); digitalWrite(in2,LOW);
                 digitalWrite(in3,HIGH); digitalWrite(in4,LOW); }
void right()    { digitalWrite(in1,HIGH); digitalWrite(in2,LOW);
                 digitalWrite(in3,LOW); digitalWrite(in4,LOW); }
void stopCar()  { digitalWrite(in1,LOW); digitalWrite(in2,LOW);
                 digitalWrite(in3,LOW); digitalWrite(in4,LOW); }

void loop() {
  if (bt.available()) {
    char cmd = bt.read();
    if (cmd == 'F') forward();
    else if (cmd == 'B') backward();
    else if (cmd == 'L') left();
    else if (cmd == 'R') right();
    else if (cmd == 'S') stopCar();
  }
}
```

  Remember: Remove HC-05 wires from Pin 0 & 1 before uploading, reconnect after!



Version 2 — ESP32 WiFi Controlled Car

Now let's upgrade! The **ESP32** has built-in WiFi, so we create a **web server** on the car itself. Open a browser on your phone → control the car! No app needed.

Smartphone Browser

ESP32 Web Server

L298N Motor Driver

Two DC Motors

ESP32 vs Arduino — Key Differences

- ESP32 has **WiFi + Bluetooth** built-in
- Faster: **240 MHz** vs 16 MHz
- More GPIO pins available
- Runs a **web server** directly
- Control via **any browser**

Wiring Changes from Version 1

- Replace Arduino Uno → **ESP32 DevKit**
- Remove HC-05 module entirely
- Motor pins same: GPIO 5, 6, 9, 10
- Power: **3.3V logic**, 7.4V battery to L298N



ESP32 creates its own WiFi hotspot — connect your phone to it!

ESP32 WiFi Car — Simplified Code Structure

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ESP32_Car";
const char* pass = "robotcar123";
WebServer server(80);

int in1=5, in2=6, in3=9, in4=10; // Same motor pins

void setup() {
  WiFi.softAP(ssid, pass);
  Serial.println("Server started!");
  // Define URL routes
  server.on("/forward", [](){ forward(); server.send(200,"text/plain","OK"); });
  server.on("/backward", [](){ backward(); server.send(200,"text/plain","OK"); });
  server.on("/left", [](){ left(); server.send(200,"text/plain","OK"); });
  server.on("/right", [](){ right(); server.send(200,"text/plain","OK"); });
  server.on("/stop", [](){ stopCar(); server.send(200,"text/plain","OK"); });
  server.begin();
}

void loop() { server.handleClient(); }

// forward(), backward(), left(), right(), stopCar() — same functions as before!
```



Assembly, Testing & Troubleshooting



Step-by-Step Assembly

O1

Mount Motors on Chassis

Fix both DC motors to the chassis using screws. Attach wheels.

O2

Connect Motors to L298N

Left motor → OUT1/OUT2. Right motor → OUT3/OUT4.

O3

Wire Microcontroller to L298N

Connect IN1-IN4 to the correct GPIO pins.

O4

Add Communication Module

HC-05 (Bluetooth) OR use ESP32 directly (WiFi).

O5

Connect Battery & Power Up

Battery → L298N 12V input. Test movement!



Common Problems & Fixes

- **Car not moving?** → Check battery voltage (should be 7V+)
- **Only one side moves?** → Check motor wire connections on L298N
- **Bluetooth not pairing?** → Reset HC-05, try password 0000
- **Motors spinning wrong way?** → Swap the two motor wires on that side
- **ESP32 not connecting?** → Forget network on phone, reconnect to "ESP32_Car"



Quick Test Checklist

- All wires secure, no loose connections
- Battery charged (use a multimeter)
- HC-05 LED blinking (pairing mode)
- Serial Monitor shows commands received
- Car responds to F, B, L, R, S
- ESP32 web server accessible at 192.168.4.1



🎉 If all checks pass — you've built your first wireless robot car!